Patrick Iben, MS

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RESEARCH ENGINEERING

Applied mathematician with 12 years in research, including seven in clinical biostatistics (nearly four as lead biostatistician on at least 15 early-phase studies) and predictive modeling of large-scale healthcare and university data. Works to eliminate needless ambiguity, risk and suffering in decisions that rest on data. Builds mathematical frameworks and turns uncertain signals into inspectable systems with traceable inputs and testable assumptions; independent work includes six manuscripts, machine learning, Python software, wearable sensing and 36 public-data consoles. Open to full-time and consulting roles.

AREAS OF EXPERTISE

Predictive Analytics | Machine Learning | Natural Language Processing (NLP) | AI/ML Model Evaluation | Mathematical and Statistical Modeling | Causal Inference | Scientific Software | Data Architecture | Data Harmonization and Quality Control | Decision-Support Systems | Clinical Biostatistics | Python | R and Shiny | SQL | SAS Programming

TECHNICAL SKILLS

Methods: predictive modeling, calibration and decision curves, causal inference with spillover and contamination, simulation, permutation inference, survival and longitudinal analysis, missing-data imputation, cluster randomized trials and estimands.

Tools: Python (PyTorch, Hugging Face Transformers, scikit-learn, XGBoost, pandas, statsmodels, FastAPI, Streamlit), R (pharmaverse and Shiny), SQL, SAS, JavaScript, Git, Docker, high-performance computing and public API integration.

Therapeutic areas: cardiovascular, dermatology, immunology, neurology, oncology, psychiatry and respiratory.

PROFESSIONAL EXPERIENCE

Biostatistician II | **Syneos Health** | Remote | December 2021 to October 2026

- Progressed from support statistician to lead biostatistician within a year; led at least 15 early-phase clinical pharmacology studies across first-in-human, dose-escalation, crossover and bioequivalence designs, and supported at least as many more.
- Initiated and created comprehensive SAS validity checks for data coding and derivations at each data stage (ADaM, SDTM and raw) of on-treatment study periods, drawing on clinical resource informatics experience; the checks were very favorably received and adopted by the internal statistical team.
- Owned protocol statistical sections, randomization schedules, statistical analysis plans and mock tables, listings and figures (TLFs); created or reviewed CDISC programming specifications for analysis datasets and outputs.
- Performed pharmacokinetic/pharmacodynamic (PK/PD) and non-compartmental analyses (NCA), wrote SAS review programs for quality control and reviewed all TLFs before delivery; contributed to clinical study reports (CSRs).
- Reviewed interim study data, annotated case report forms (CRFs) and database designs against analysis requirements.

Associate Biostatistician | **Emmes** | Rockville, MD | September 2019 to December 2021

- Worked on opioid-use-disorder studies for the National Institute on Drug Abuse (NIDA) with nodes of its Clinical Trials Network; wrote protocol statistical sections, analysis plans and the statistical sections of final study and Data and Safety Monitoring Board (DSMB) reports.
- Created a framework for auto-coding MedDRA and WHODrug codes from manually entered verbatim fields, using natural language processing (NLP) and machine learning.
- Developed dynamic progress reporting that gave teams timely visibility, plus SAS/R data-management and analysis tools.
- Harmonized electronic health record (EHR) data from four rural clinics for an opioid-use-disorder study and authored the extraction and normalization programs, credited by name in a 2024 *Addiction* paper; supported external research partners.

Research Assistant and Teaching Assistant | **University of Colorado Anschutz** | Aurora, CO | August 2016 to July 2019

- Constructed, harmonized and maintained multi-state hospital EHR databases with millions of records and hundreds of derived variables as biostatistics research assistant for the Data to Value nursing econometrics initiative at the University of Colorado College of Nursing, supporting pediatric nursing-cost research and reuse in later studies.
- Advised about three doctoral nursing candidates on study design and calculated their sample sizes.
- Served for three years as teaching assistant for multiple biostatistics courses, including Biostatistical Methods and Advanced Data Analysis, with office hours, review sessions, occasional lectures and exam proctoring.

Earlier roles | 2014 to 2016

- **Postsecondary and Policy Analysis Fellow, State Higher Education Executive Officers.** Prepared analyses for a Gates Foundation-funded national assessment of student learning outcomes; gave statistical support on a federal grant proposal.
- **Data Analysis Fellow, Yale School of Child Development.** Evaluated assistive technology for students with print disabilities.
- **Head Undergraduate Research Assistant and Math Tutor, Southern Connecticut State University.** Harmonized and analyzed 10 years of university cohort data and built predictive models of academic success and retention; designed student survey questions later validated as predictors; tutored undergraduates in statistics, calculus and algebra for a year.

SELECTED PROJECTS

Trial biostatistics toolkit and eCRF pipeline apps

A public toolkit for biostatisticians: SAS and R templates, dashboards and quality-control checks from protocol statistical sections to report text. It runs locally, so client data stay on the analyst's machine; the example studies are synthetic.

- Created a worked example of a bioequivalence study: one synthetic trial from sample size and randomization through the analysis plan, analysis datasets, pharmacokinetic analysis, tables and report text, all regenerated by a single script.
- Designed safety-monitoring tools for first-in-human studies: liver and cardiac screening rules, a Safety Review Committee packet and an interactive dashboard that runs in the browser to streamline safety review.
- Engineered a quality-control harness that recomputes key results a second way and is tested against planted errors.
- Wrote SAS and R program libraries for standard tables, listings and figures across common early-phase designs, plus simulation studies of two regulatory checks.
- Released four Streamlit apps for reviewing electronic case report form (eCRF) data (medical coding, de-identification, participant risk and central monitoring), trained XGBoost models for the first two on synthetic data and published their measured limitations.

Public Data Consoles and Coastal Water Resilience

- Built 36 browser-only decision-support consoles on public data feeds and dated snapshots across hazards, public health, access to care and markets; each labels every input as measured, modeled or demo so users can judge evidence quality.
- Unified two original research methods in Coastal Water Resilience: a ten-metric diagnostic suite from contamination-resilient cluster randomized trial design plus directed source/sink topology accounting, applied to live NOAA and National Weather Service forecasts across 131 published wastewater lift stations to drive projected impacts and response plans.

Autonomic Telemetry and Resilience Telemetry

- Prototyped a personal physiological-arousal biofeedback system combining heart rate, heart-rate variability, EEG and covariance distance against a personal reference, on live Muse S and Hume Band streams via a local Mac-to-iPhone bridge.
- Applied original directed-contagion research to cross-asset market volatility: spillover networks, transmitter stability and historical replay, with held-out tests of predictive value against baselines that track the loudest and most persistent assets.

Consilience Engine

- Packaged gene-set analysis as a Python library and command-line tool using InterPro, Human Protein Atlas, STRING and Gene Ontology; evaluated with permutation tests on a 410-gene nervous-system map and a 132-gene immune replication.

catmech R package

- Submitted catmech to CRAN in November 2026, an R package for regression when only a subset of records has been checked against a reference.

EDUCATION AND TRAINING

MS, Biostatistics, Data Science Analytics minor | University of Colorado Anschutz
BS, Mathematics, Applications of Mathematics | Southern Connecticut State University | Magna cum laude
High-Performance Computing Training and Certification | University of Colorado Anschutz
Diploma in Audio Production Technologies | School of Audio Engineering

PUBLICATIONS AND MANUSCRIPTS

Submitted manuscripts, with review status as of 1 November 2026:

- **Contamination-Resilient Cluster Randomized Trial Design under Spatial Interference: A Simulation Study of Contamination Control.** *Statistics in Medicine*, submitted 19 October 2026. Began as master's research at the University of Colorado and resumed years later. Preprint doi:10.5281/zenodo.23486176.
- **Auditing preprocessing-sensitive allele rankings in predicted chromatin contact maps.** *BMC Bioinformatics*, submitted 29 October 2026. Integrates an earlier preprint, doi:10.5281/zenodo.23543154.
- **A false floor: Refuting our own information-limit claim in a lethal identification task.** *PLOS ONE*, submitted 22 October 2026. Preprint doi:10.5281/zenodo.23519299.
- **Anticipating Network Contagions: When Directedness Outperforms Connectedness.** *Applied Network Science*, with editor.
- **Which data modality recognizes a gene set? A domain-agnostic consilience engine.** *PeerJ*, in peer review.
- **No emotion-specific image-computable code in the human left amygdala.** *Peer Community In*, preprint review. Preprint doi:10.5281/zenodo.21422754.

Published work:

- **Acknowledged contributor:** Osterhage KP, Hser YI, Mooney LJ, et al. Identifying patients with opioid use disorder using International Classification of Diseases (ICD) codes: Challenges and opportunities. *Addiction*. 2024;119(1):160-168.
- **Co-author:** Ben-Avie M, Rosenwald L, Depastino R, Iben P. How can we help young adults with autism thrive in the workplace? *The Conversation*, 2016.